Simon Donker van Heel

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Contact Information

Econometric Institute, Erasmus University Rotterdam
Burgemeester Oudlaan 50, 3062 PA Rotterdam, the Netherlands

Research Fields

Primary: Econometrics; Time Series Analysis
Secondary: Financial Econometrics; Statistical Methodology

Education

Erasmus University Rotterdam and Tinbergen Institute 2023 – 2027
PhD in Econometrics, expected completion August 2027
Thesis: Time series filtering under misspecification
Supervisors: Prof. Dick van Dijk and Dr. Rutger-Jan Lange

Research visit: Department of Statistics, Harvard University Fall 2025
Fulbright Scholar; host: Prof. Neil Shephard

Tinbergen Institute 2021 – 2023
MPhil in Economics, Advanced Econometrics track

Vrije Universiteit Amsterdam
MSc in Econometrics and Operations Research, *cum laude* 2020 – 2021
BSc in Business Analytics 2016 – 2020
Exchange semester: Nanyang Technological University, Singapore 2019 – 2020

Job Market Paper

"Filtering without recursion and some of its uses in financial economics" (2026), with Neil Shephard.
arXiv | Tinbergen Institute Discussion Paper | SSRN | Web appendix

Working Papers

"Gradient-based filtering under misspecification: Stability and error bounds" (2025), with Rutger-Jan Lange, Bram van Os, and Dick van Dijk.
Major revise and resubmit, Journal of Econometrics.
arXiv | Tinbergen Institute Discussion Paper | SSRN

"Exponentially weighted estimands and the exponential family: Filtering, prediction and smoothing" (2025), with Neil Shephard.
Reject and resubmit, Journal of Econometrics.
arXiv | Tinbergen Institute Discussion Paper

Work in Progress

"An implicit dynamic multiple quantile filter"
Draft available on request.

"Implicit score-driven high-dimensional precision matrices",
with Rutger-Jan Lange, Bram van Os, and Dick van Dijk.

Grants and Awards

Runner-up Best Poster Award, Netherlands Econometric Study Group 2025, 2026
Fulbright Scholarship, research visit to Harvard University, $8,000 2025
Cultuurfonds grant, €8,740 2025
Ketel 1 Fonds grant, €3,000 2025
Erasmus Research Institute of Management (ERIM) grant, €3,000 2025
Erasmus Trustfonds grant, €2,000 2025
Second place (of 31 universities, as team captain), Econometric Game 2025
Best Poster Award, Netherlands Econometric Study Group 2024
I.S.E.O. Summer School scholarship, €1,600 2024
Tinbergen Institute scholarships and tuition waivers, around €20,000 2021 – 2023

Presentations

Tinbergen Institute PhD Seminar, Amsterdam 2026
Brown Bag Seminar, Department of Econometrics and Data Science, Vrije Universiteit Amsterdam
Econometric Institute Seminar, Erasmus University Rotterdam
Symposium on Econometric Theory and Applications (SETA), University of Tokyo
Netherlands Econometric Study Group, Tilburg (poster)
Score-Driven and Nonlinear Time Series Models, Ca' Foscari University of Venice
PhD Internal Conference (EPIC), Erasmus University Rotterdam

Statistics Department Seminar, Harvard University 2025
NBER-NSF Time Series Conference, Rutgers University, New Brunswick (poster)
Netherlands Econometric Study Group, Leuven (poster)

International Association for Applied Econometrics (IAAE), Thessaloniki 2024
European Economic Association – European Meeting of the Econometric Society (EEA-ESEM), Rotterdam
I.S.E.O. Summer School, Milan (poster)
Netherlands Econometric Study Group, Maastricht (poster)
New York Camp Econometrics, Lake Placid (poster)
Dynamic Econometrics Conference, University of Oxford
Tinbergen Institute PhD Seminar, Amsterdam
PhD Internal Conference (EPIC), Erasmus University Rotterdam

Teaching Experience

Erasmus University Rotterdam

Seminar Case Studies in Financial and Applied Econometrics (MSc) 2025 – present
10-week research projects for companies, in teams of 4. Supervised 4 teams on GDP nowcasting for Ortec Finance in 2026, with weekly meetings, feedback on drafts and grading.

Quantitative Risk Management (MSc) 2023 – present
Graded the written exam and the resit.

Financial Econometrics Case Study (BSc) 2023 – 2025
2nd-year undergraduate course on forecasting stock market volatility. Held meetings with the teams and graded the reports.

Academic Skills (BSc) 2023 – 2025
Academic writing and presenting for 2nd-year econometrics students. Taught tutorials, gave feedback on reports and graded the presentations.

Thesis supervision (BSc and MSc) 2023 – present
5 BSc theses and 1 MSc thesis per year. Second assessor for 3 MSc theses per year.

Other Experience

Private Equity Intern, Nordian Capital Partners, Amsterdam 2020

Personal Information

Citizenship: Dutch
Languages: Dutch (native), English (fluent)
Programming: MATLAB (preferred), R, Python
Interests: arthouse films, reading, padel, travelling, skiing

References

Prof. Dick van Dijk, Erasmus University Rotterdam, djvandijk@ese.eur.nl
Dr. Rutger-Jan Lange, Erasmus University Rotterdam, lange@ese.eur.nl
Prof. Neil Shephard, Harvard University, shephard@fas.harvard.edu
Prof. Siem Jan Koopman, Vrije Universiteit Amsterdam, s.j.koopman@vu.nl